

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

- | n/a | Confirmed |
|-------------------------------------|--|
| ☐ | ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☐ | ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☐ | ☒ The statistical test(s) used AND whether they are one- or two-sided
<i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i> |
| ☒ | ☐ A description of all covariates tested |
| ☐ | ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☒ | ☐ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

- | | |
|-----------------|---|
| Data collection | For indirect immunofluorescence assay (cell line) and average infection ratio were measured by Celigo Image Cytometer (Nexcelom Bioscience).
The numbers of SARS-CoV-2, OC43, 229E, IAV foci were calculated using an EliSpot reader (Cellular Technology Ltd.). |
| Data analysis | H&E image were process by CaseViewer_2.4 software.
Raw data were analyzed using Microsoft Excel software(Microsoft 365), GraphPad PRISM (Versions 9) |

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The atomic coordinates included in this study have been deposited in the PDB with the accession codes: 8WTS. Source data are provided with this paper. All relevant data are available in the article, supplementary Information, or from the corresponding author J.Z. upon reasonable request.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	Participants were not excluded based on sex or gender.
Reporting on race, ethnicity, or other socially relevant groupings	No socially constructed or socially relevant categorization variables were used in this manuscript.
Population characteristics	Population characteristics are described in Supplementary table 1.
Recruitment	In this study, patients were recruited with PCR-confirmed COVID-19. Participants were recruited from a hospital located in Guangzhou, Guangdong, China, named the First Affiliated Hospital of Guangzhou Medical University from January 2020 to January 2023. Informed consent was obtained from all participants prior to sample collection. Bronchoalveolar lavage fluid (BALF) samples and peripheral blood mononuclear cells (PBMCs) were obtained by trained clinicians from two cohorts: healthy donors and individuals who had tested positive for SARS-CoV-2 and diagnosed with COVID-19. All methods were performed according to the relevant guidelines and regulations.
Ethics oversight	All human subjects research was approved by the First Affiliated Hospital of Guangzhou Medical University (Ethics ID: ES-2020-65, ES-2023-015-01). All study participants have provided written informed consents and did not received compensation for taking part in this study.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	In Fig 3 and Fig 6, for SARS-CoV-2 and IAV infection, six- to eight-week old C57BL/6 mice were infected. From 4 to 12 mice per group in our study and displayed the number in the figures. In Fig 3 and Extended Data Fig 1, for SARS-CoV-2 infection, six- to eight-week old BALB/c mice were infected, with 3 mice per group. The determination on the sample sizes in in vivo study is based on the expected variations in the disease model and 3R principles (Replacement, Reduction and Refinement). In vitro: no special considerations
Data exclusions	No data were excluded
Replication	All experiments were repeated in at least duplicates except indicated. Similar findings were obtained from all repeats.
Randomization	There was no allocation except for the grouping of the animals. Animals were randomly allocated to the groups.
Blinding	No blinding was done. Blinding was not relevant to the study because the results are quantitative and objective, and does not require a subjective judgment.

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	
Research sample	
Sampling strategy	
Data collection	
Timing	
Data exclusions	
Non-participation	
Randomization	

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	
Research sample	
Sampling strategy	
Data collection	
Timing and spatial scale	
Data exclusions	
Reproducibility	
Randomization	
Blinding	

Did the study involve field work? ☐ Yes ☐ No

Field work, collection and transport

Field conditions	
Location	
Access & import/export	
Disturbance	

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

SARS-CoV / SARS-CoV-2 Nucleocapsid Antibody (Cat. No.: 40143-R004, Sino Biological, Inc. Beijing)
 SARS-CoV-2 Nucleocapsid Antibody, Rabbit
 HCoV-229E Nucleocapsid Antibody, Rabbit PAb (Cat. No.: 40640-T62, Sino Biological, Inc. Beijing)
 HCoV-OC43 Nucleocapsid Antibody, Rabbit PAb (Cat. No.: 40643-T62, Sino Biological, Inc. Beijing)
 Pan Influenza A Nucleoprotein Antibody, Rabbit mAb (Cat. No.: 40208-R010, Sino Biological, Inc. Beijing)
 HRP-labelled goat anti-rabbit secondary antibody (Cat. No.: 111-035-144, Jackson ImmunoResearch Laboratories)
 Goat anti-Rabbit IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor™ Plus 488 (Jackson ImmunoResearch Laboratories, 711-545-152, Inc. West Grove, PA)
 ACE2 Polyclonal Antibody, Rabbit PAb (Cat. NO.:21115-1-AP, Proteintech, Inc. Wuhan)
 SARS-CoV-2 (2019-nCoV) Spike S2 Antibody, Rabbit PAb (Cat. NO. 40590-T62, Sino Biological, Inc. Beijing)
 HRP-conjugated GAPDH Antibody, Mouse mAb (Cat. NO. AC035, ABclonal, Inc. Wuhan)
 HRP-conjugated Goat Anti-Mouse IgG (H+L) Secondary Antibody, Goat mAb (Cat. NO. SA00001-1, Proteintech, Inc. Wuhan)
 HRP-conjugated Goat Anti-Rabbit IgG (H+L) Secondary Antibody, Goat mAb (Cat. NO. SA00001-2, Proteintech, Inc. Wuhan)
 APC anti-HA.11 Epitope Tag Antibody, Mouse mAb (Cat. NO. 901523, Biolegend)
 All primary antibodies were used at a dilution of 1:1000 for immunoblotting, 1:2000 for immunofluorescence and focus forming assay
 All secondary antibodies were used at a dilution of 1:2500 or 1:5000 for immunoblotting, 1:1000 for immunofluorescence, 1:5000 for focus forming assay.

Validation

All primary and secondary antibodies used are commercial antibodies reported by the manufacturer to be validated for use

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)

African green monkey kidney Vero cell (Vero E6)(CRL-1586), THP-1 (TIB-202), Calu3 (HTB-55), A549 (CRM-CCL-185), MDCK (CCL-34), 293T (CRL-3216), Caco2 (HTB-37), and HRT-18 (CCL-244) cells were obtained from ATCC. Huh7 (JCRB0403) cells was obtained from JCRB Cell Bank.

Authentication

All cell lines were frequently checked for cellular morphologies, growth rates and functions. All purchased cell lines were available in commercial company.

Mycoplasma contamination

All cell lines used were tested (by PCR) and were mycoplasma free.

Commonly misidentified lines
(See [ICLAC](#) register)

No commonly misidentified lines were used in this study.

Palaeontology and Archaeology

Specimen provenance

Specimen deposition

Dating methods

☐ Tick this box to confirm that the raw and calibrated dates are available in the paper or in Supplementary Information.

Ethics oversight

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Animals and other research organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals	For SARS-CoV-2 and IAV infection, specific pathogen-free 6-8-week-old WT C57BL/6 mice were purchased from GemPharmatech Co., Ltd. (Jiangsu, China). GALNT2 KO mice were purchased from Gempharmatech Co.. IFNAR-/- BALB/c mice were generated by backcrossing IFNAR-/- BALB/c with WT BALB/c mice for at least ten generations. STAT1-/- mice were purchased from Jackson Laboratories.
Wild animals	No wild animals were used in the study.
Reporting on sex	
Field-collected samples	No field collected samples were used in the study.
Ethics oversight	All animal experiments were complied with relevant ethical regulations regarding animal research. The animal study was reviewed and approved by Institutional Animal Care and Use Committees of the First Affiliated Hospital of Guangzhou Medical University (2022114) and Guangzhou Customs Inspection and Quarantine Technology Center (IQC202306).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Clinical data

Policy information about [clinical studies](#)

All manuscripts should comply with the ICMJE [guidelines for publication of clinical research](#) and a completed [CONSORT checklist](#) must be included with all submissions.

Clinical trial registration	
Study protocol	
Data collection	
Outcomes	

Dual use research of concern

Policy information about [dual use research of concern](#)

Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

No	Yes	
☐	☐	Public health
☐	☐	National security
☐	☐	Crops and/or livestock
☐	☐	Ecosystems
☐	☐	Any other significant area

Experiments of concern

Does the work involve any of these experiments of concern:

No	Yes
☐	☐ Demonstrate how to render a vaccine ineffective
☐	☐ Confer resistance to therapeutically useful antibiotics or antiviral agents
☐	☐ Enhance the virulence of a pathogen or render a nonpathogen virulent
☐	☐ Increase transmissibility of a pathogen
☐	☐ Alter the host range of a pathogen
☐	☐ Enable evasion of diagnostic/detection modalities
☐	☐ Enable the weaponization of a biological agent or toxin
☐	☐ Any other potentially harmful combination of experiments and agents

Plants

Seed stocks	no
Novel plant genotypes	no
Authentication	no

ChIP-seq

Data deposition

- ☐ Confirm that both raw and final processed data have been deposited in a public database such as [GEO](#).
- ☐ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links <i>May remain private before publication.</i>	
Files in database submission	
Genome browser session (e.g. UCSC)	

Methodology

Replicates	
Sequencing depth	
Antibodies	
Peak calling parameters	
Data quality	
Software	

Flow Cytometry

Plots

Confirm that:

- ☐ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☐ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☐ All plots are contour plots with outliers or pseudocolor plots.
- ☐ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation

Instrument

Software

Cell population abundance

Gating strategy

- ☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.

Magnetic resonance imaging

Experimental design

Design type

Design specifications

Behavioral performance measures

Acquisition

Imaging type(s)

Field strength

Sequence & imaging parameters

Area of acquisition

Diffusion MRI

☐

Used

☐

Not used

Preprocessing

Preprocessing software

Normalization

Normalization template

Noise and artifact removal

Volume censoring

Statistical modeling & inference

Model type and settings

Effect(s) tested

Specify type of analysis: ☐ Whole brain ☐ ROI-based ☐ Both

Statistic type for inference

(See [Eklund et al. 2016](#))

Correction

Models & analysis

n/a

Involvement in the study

☐

☐

Functional and/or effective connectivity

☐

☐

Graph analysis

☐

☐

Multivariate modeling or predictive analysis

Functional and/or effective connectivity

Graph analysis

Multivariate modeling and predictive analysis